Tobe Energy — Techno-Economic Analysis

Investment-Grade Summary | March 2026

The Opportunity

The Problem No One's Solved

Grey hydrogen costs $2/kg to produce.

But you pay $15–25/kg delivered.

85% of what you're paying isn't hydrogen. It's logistics.

Grey hydrogen only makes economic sense when manufactured at enormous scale in centralized hubs — Gulf Coast, Alberta, the same facilities that have dominated the industry for 50 years. Every industrial customer more than 100 miles from a hub is captive to trucking economics.

94 million tonnes per year. $226 billion in annual demand. All locked into a model where the product is cheap but moving it is ruinous.

What We Built

Tobe Energy produces hydrogen on-site using membrane-free electrolysis with pulsed waveform architecture.

- **No precious metals.** No membranes. No rare earths.
- **Near-ambient operation.** No chillers, no thermal management.
- **>92% system efficiency (HHV).** Up to 94.7% at the stack.
- **First electrolyzer under DOE's $2/kg target** at industrial electricity rates.

We don't ship hydrogen. We ship machines that make it.

The Economics

Anchor Case: Repeat-Commercial Facility

Single facility, third deployment. Proven design, no pilot risk.

Metric	Value
Hydrogen Price	$25.00/kg delivered (regional avg)
Production Cost	$2.62/kg (electric + compression + water)
Gross Margin	89.5%
Project IRR	217.7%
10-Year NPV	$57.5M
Payback	<18 months

Why margins are structural, not temporary: - No transport cost — produced where consumed - No storage cost — produced on demand - No compression to DOT transport spec — only to customer use pressure - Scale-independent — unit economics work at 125 kg/day

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Seven-Year Company Projection

Per FINAL_Tobe_Seed_Financial_Model_v8.5.1

Revenue Build

Fiscal Year	H_2 Revenue	Equipment	Services	45V PTC	Total
FY1 (Apr '26–Mar '27)	$0	$0.9M	$0	$0	$0.9M
FY2	$12.4M	$4.3M	$0.1M	$1.1M	$17.8M
FY3	$37.1M	$10.7M	$0.5M	$3.7M	$52.0M
FY4	$78.8M	$22.7M	$1.4M	$7.9M	$110.8M
FY5	$138.1M	$39.7M	$2.9M	$12.5M	$193.2M
FY6	$214.7M	$52.9M	$5.0M	$10.8M	$283.5M
FY7	$270.0M	$51.1M	$7.6M	$10.8M	$339.4M

Margin Profile

Metric	FY3	FY5	FY7
Gross Margin	83.0%	82.5%	85.0%
EBITDA Margin	69.0%	67.5%	69.0%
Net Margin	48.0%	49.0%	47.0%

Note: EBITDA turns positive in FY2 at $6.3M and scales to $234.9M by FY7.

Capital Structure

Fundraising Trajectory

Round	Timing	Amount	Post-Money	Primary Use
Pre-Seed	Complete	$2.0M	—	Prototype, team
Seed	**Now**	**$10.0M**	$45M	First MW deployment, UL certification
Series A	Q2 2027	$57.0M	$200M	Manufacturing scale-up, first replication
Series B	Q1 2029	$60.0M	$425M	Geographic expansion
Series C	Q4 2030	$225.0M	$1.1B	National footprint

Seed Allocation ($10M): - R&D & Engineering: $1.1M (T-125 completion, UL certification) - Manufacturing Equipment: $1.1M (production line tooling) - Personnel: $2.2M (engineering, manufacturing crew) - Zeeco Deployment: $2.5M (facility buildout, first MW) - Working Capital & Contingency: $3.1M

Debt Strategy

$2.5M venture debt at Seed (CSC Leasing terms, 7% rate, 36-month amortization). Project financing unlocks post-Series A for expansion facilities.

Product Line

T-25 | Entry Product

- **Output:** 25 kg/day
- **Nameplate:** 50 kW

- **Base Price:** $25,000
- **Optioned:** $35,000 (remote monitoring, SCADA, consumables, extended warranty)
- **Target Market:** Industrial gas distributors, fleet depots, research institutions

T-125 | Commercial Workhorse

- **Output:** 125 kg/day
- **Nameplate:** 250 kW
- **Base Price:** $120,000
- **Optioned:** $167,000
- **Target Market:** Manufacturing facilities, food processing, ammonia producers

T-2500 | Utility Scale (In Development)

- **Output:** 2,500 kg/day (20x T-125)
- **Nameplate:** 5 MW
- **Target Price:** $3,000,000
- **R&D Timeline:** Design complete FY3, production FY4
- **Target Market:** Refineries, chemical plants, large industrial consumers

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Unit Economics at Scale

Per-Facility Economics (T-125 deployment, FY3+)

Line Item	$/kg	Notes
Electricity	$1.47	42 kWh/kg @ $0.035/kWh
Compression	$0.17	3.5 kWh/kg to 350 bar
Transport	$0.91	50-mile delivery (owned logistics)
Water	$0.05	9L/kg at RO rates
Direct OpEx	**$2.62/kg**	
Sale Price	$25.00/kg	Regional delivered avg
Gross Profit	**$22.38/kg**	**89.5% margin**

Manufacturing Cost-Down Trajectory

Phase	Cumulative Units	T-125 BOM	$/kW	Gross Margin
Launch (Y2)	10	$100,000	$400	40.1%
Volume (Y3)	40	$96,000	$384	42.5%
Optimized (Y4)	100	$92,500	$370	44.6%
Mature (Y5)	200	$87,500	$350	47.6%
Scale (Y7)	520	$77,500	$310	53.6%

Learning Rate: ~5% compound annual BOM reduction on T-125.

45V Production Tax Credit

The Inflation Reduction Act's 45V PTC provides $3.00/kg for clean hydrogen meeting lifecycle emissions thresholds.

Facility Eligibility

Facility	Construction Start	45V Eligible
F1 Tulsa	Apr 2026	✅ Yes
F2 OKC	Oct 2026	✅ Yes
F3 Seattle	Apr 2027	✅ Yes
F4 Spokane	Sep 2027	✅ Yes
F5 Albuquerque	Feb 2028	❌ No
F6+	Dec 2028+	❌ No

10-Year PTC Value per Eligible Facility: $27M ($3/kg × 900,000 kg/year × 10 years)

Note: Construction must begin before December 31, 2027 per IRS guidance. Four facilities qualify under current timeline.

Key Risks & Mitigations

Technology Risk

Status: Proven at prototype scale. T-25 in production. T-125 at Zeeco ARC commissioning Q2 2026.

Mitigation: Third-party efficiency validation scheduled. UL 2264A certification in progress (targeting Q4 2026 — Ohmium was first certified Nov 2024, demonstrating clear pathway).

Operational Risk

Status: First commercial deployment. Limited operating history at scale.

Mitigation: Zeeco partnership provides ARC facility infrastructure, operational support, and reference customer. EPC capability de-risks deployment execution.

Maintenance & Durability

Status: Long-duration testing ongoing. Modeled at 5% annual replacement reserve.

Mitigation: Membrane-free design eliminates primary degradation mechanism of PEM systems. No precious metal catalyst degradation.

Market Risk

Status: $100M pipeline. First deployment contracted.

Mitigation: Pricing at $25/kg positions significantly below grey hydrogen delivered cost ($15–25) while maintaining 70%+ gross margins. Room to compete on price if needed.

Regulatory Risk

Status: 45V guidance clarifying. BABA compliance achieved.

Mitigation: 95% US-sourced content. Oklahoma manufacturing provides political and supply chain resilience.

Data Room Requirements (Per Editorial Companion)

The following items are prioritized for diligence support:

Available Now

- ✅ Pulse Holdings Quote TE-PH-ELX-88-51 (equipment pricing)
- ✅ Internal efficiency test data (stack-level)
- ✅ Zeeco ARC contract (redacted)
- ✅ December 2025 financials ($1.13M cash, $109K/mo burn)

In Progress

- 🔄 Third-party efficiency validation (pending independent test)
- 🔄 UL 2264A certification package
- 🔄 Long-duration reliability data (ongoing)

Customer Evidence (Sanitized)

- 🔄 Anonymized LOI summary (pipeline documentation)
- 🔄 Pricing appendix supporting $25/kg regional delivered rate

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Why This Works

1. Grey hydrogen's business model is broken. 85% of delivered cost is logistics. The entire industry is built on moving a commodity that's cheap to make but expensive to transport.

2. We eliminated the cost stack's biggest line item. On-site production. No trucking. No storage terminals. No hazmat logistics.

3. The margins are structural. At $2.62/kg production cost and $25/kg delivered pricing, we're not competing on efficiency improvements. We're competing on business model.

4. The market isn't going anywhere. 94M tonnes/year of required industrial demand. Refining, ammonia, steel. These aren't discretionary. They're locked in.

5. The timing is now. 45V provides $3/kg tailwind for the next decade. Facilities built before 2028 capture $27M+ per site in tax credits.

The Ask

$10M Seed Round

Deploying capital to: - Complete first MW commercial installation at Zeeco ARC - Achieve UL certification for US market access - Scale manufacturing for T-125 production - Build to 15-person team for Series A execution

Use of funds detailed in financial model. 18-month runway to Series A milestones.

Tobe Energy gives the present a choice. And the future, a chance.

Document Version

TEA Narrative v1.0 | March 2026 Financial Model Reference: v8.5.1 Editorial Companion Reference: March 2026

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